



Cluster : Cyber Forensics

Problem Statement title : Artificial Intelligence based Log Investigation Framework for Next-Generation Cyber Forensics

Description:

With the exponential growth of cybercrimes and the proliferation of devices, massive amounts of logs are generated from computers, mobile devices, IoT, servers, and applications. These logs are crucial for reconstructing timelines, understanding the nature of incidents, and drawing actionable insights. Current manual or semi-automated methods are time-consuming and inefficient. There is a pressing need for an AI-powered framework that can ingest, parse, and analyze logs, enabling investigators to correlate events, detect anomalies, and generate actionable forensic insights efficiently.

Exact Deliverables :

- Web-Based Dashboard – Interactive and user-friendly interface for managing and investigating logs.
- Log Ingestion & Parsing Module – Capable of handling diverse log formats across OS, devices, and applications.
- Database Storage Layer – Efficient preservation and retrieval of ingested logs.
- AI Correlation & Inference Engine – Machine learning models for log correlation, anomaly detection, and inference.
- Filtering & Search Tools – Advanced queries and filters to aid investigation.
- Automated Reporting – Exportable reports in PDF, CSV, JSON formats.
- (Optional) LLM Prompt Integration – Natural language queries for ease of interaction.

Milestones, Evolution Parameters:

- Phase 1: Implement log ingestion, parsing, and aggregation with basic filtering/search.
- Phase 2: Enable secure data storage/retrieval and prototype AI engine for anomaly detection.
- Phase 3: Build graphical representation and intuitive GUI.
- Phase 4: Deliver full-featured reporting engine and conduct benchmarking.

Additional Information:

- Data privacy and chain of custody must be preserved, ensuring logs remain tamper-proof.
- Explore explainable AI (XAI) techniques so investigators can understand AI-driven correlations.
- Consider cloud deployment models for scalability and integration with SIEM/SOAR ecosystems.